

EX ANTE ANALYSIS · FULL WALKTHROUGH

Mapping the public and private returns to Enhanced Rock Weathering in Sub-Saharan Africa

A spatially explicit, decomposed profitability model that asks where each return justifies deployment, and where **neither suffices alone yet together they do.**

44

SSA countries on a per-pixel grid

23

SPAM crops, area-weighted to each pixel

~0.083°

grid resolution of the decision object

2.7 Mha

combined-only band, the policy object

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METHOD

Decomposed per-pixel gross margin

HEADLINE REGIME

Equilibrium, net-export · targeted / uniform-20

STATUS

Working draft · Agricultural Economics · 2026-09 run

THE PROPOSITION

ERW pays the farmer twice.

Spreading finely ground basalt on cropland couples two value streams that are usually separate:

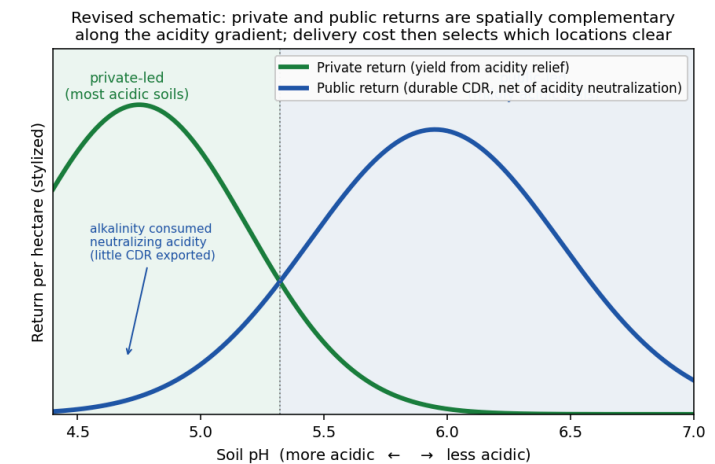
PRIVATE · AGRONOMIC

Alkalinity neutralizes soil acidity like agricultural lime, raising yields. Accrues to the farmer.

PUBLIC · CARBON (CDR)

Weathering durably removes atmospheric CO₂, monetizable through voluntary carbon markets (VCM).

Central question: where does each return justify deployment, and where does neither suffice alone, yet together they do?



Realized returns are additionally gated by the delivery-cost gradient (distance to basalt): within each band, only low-cost, basalt-proximate pixels clear.

WHAT WE ASK

Six research questions.

Q1

Where does the **private** (yield) return alone justify ERW investment?

Q2

Where does the **public** (carbon) return alone justify it?

Q3

Where do the two **coincide**, the intersection?

Q4

Where does neither stream alone suffice, but the **combined margin** is positive?

Q5

How **robust** is the deployable geography to parameter assumptions?

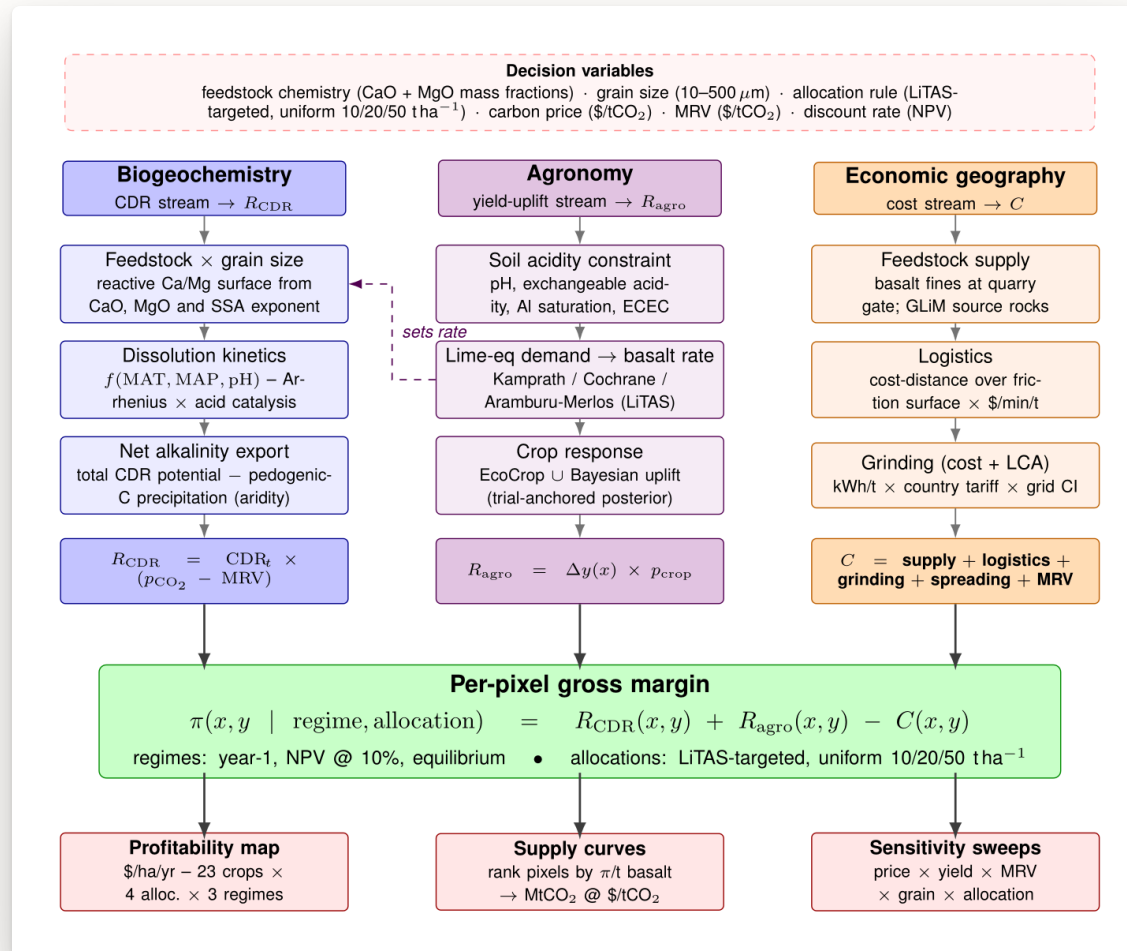
Q6

What would shift the **public (carbon) envelope** outward?

Q4 is the central result; Q5 and Q6 define the robust core and the levers on the carbon case.

MODELING FRAMEWORK

A decomposed per-pixel gross margin.



$$\pi = R_{\text{agro}} + R_{\text{CDR}} - C$$

- **Three streams.** Agronomy (yield uplift), biogeochemistry (durable CDR), and a fully decomposed delivered cost.
- **Resolution.** $\sim 0.083^\circ$ grid $\times$ 44 SSA countries $\times$ 23 SPAM crops, area-weighted to each pixel.
- **Decision object.** Each pixel is classified by which stream(s) make the margin positive, not a single supply curve.
- **Regimes.** Year-1, NPV @ 10%, and equilibrium steady state (headline).

INPUTS

Public data layers behind every pixel.

LAYER	SOURCE	WHAT IT SUPPLIES
Soil	SoilGrids	pH, ECEC, texture on cropland
Crops	SPAM v2	23 crop distributions & production
Agronomy	EcoCrop	crop pH-suitability response
Feedstock	GLiM	basalt source geology
Logistics	MAP friction	travel-time delivered-cost surface
Climate	WorldClim	temperature & moisture for weathering
Prices	FAOSTAT	2016–20 producer-price medians
Lime req.	LiTAS	lime-equivalent dose (targeting)
Carbonate	Merlos / Kamprath	acidity sink for net-export

PART I

The assumption sets

Five modeling choices govern every result: delivered cost, carbon price, net-export CDR accounting, an over-liming private penalty, and kinetic weathering saturation.

ASSUMPTION 1 · DELIVERED COST

Cost is dominated by transport, a real lever.

DECOMPOSED COST

Grinding + feedstock + freight + spreading + MRV, built up per pixel from the friction surface.

TRANSPORT BINDS

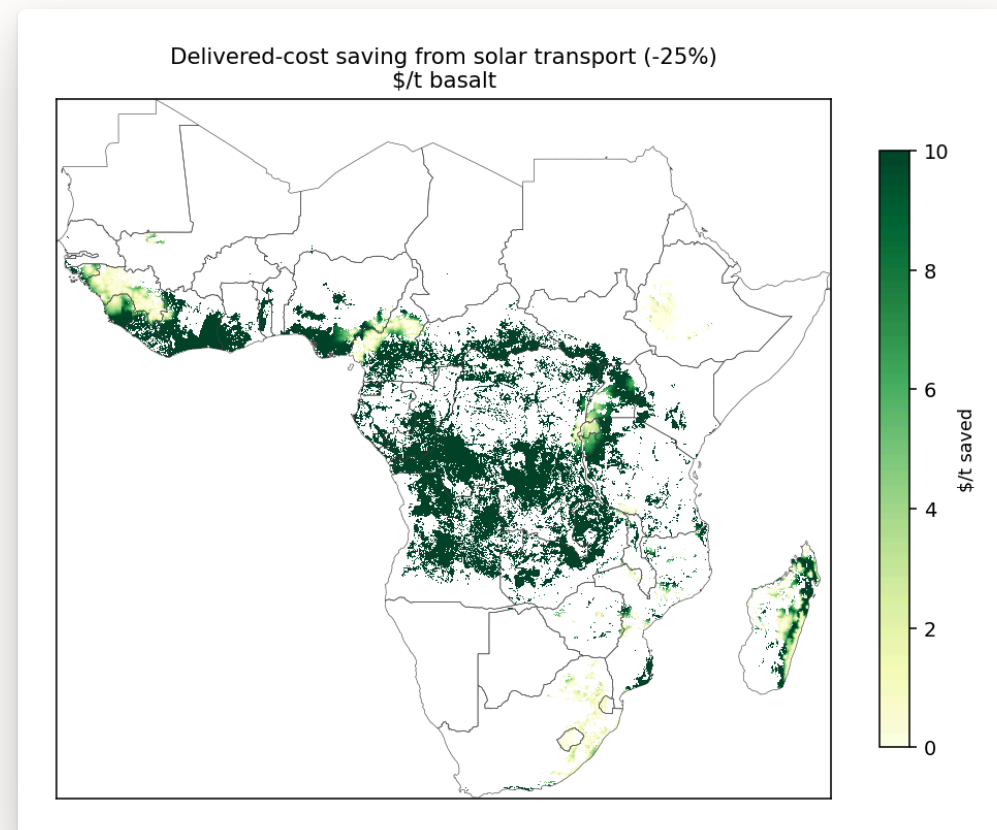
Delivered cost rises steeply with travel time; remote acid soils are priced out even where agronomy is strong.

THE SOLAR LEVER

Electrified / solar-assisted haulage cuts variable transport cost; a ~25–40% reduction visibly expands the viable frontier.

MRV

A per-tonne monitoring cost m is netted from the carbon price (see Assumption 2).



ASSUMPTION 1 · DELIVERED COST

Delivered basalt cost, decomposed.

COMPONENT	COST PER T BASALT	TYPE	BASIS / SOURCE
Quarry-gate feedstock	\$10 / t	fixed	basalt fines (aggregate by-product)
Grinding (50 μm)	19.4 kWh/t × local tariff (≈\$1.9/t)	fixed *	Strefler et al. 2018 energy curve
Transport (haulage)	travel-time × \$0.04/min/t	variable	30 t truck over MAP friction surface
Spreading	\$8 / t	fixed	on-farm application
Delivered cost / t	quarry + grinding + transport + spreading	= sum	the delivered \$/t consumed per pixel
MRV (netted separately)	\$20 / tCO ₂	per tCO ₂	netted from carbon price; low end of the \$15–71/tCO ₂ range (Mercer et al. 2024)

* Grinding cost varies with the country electricity tariff. Transport is the only spatially varying term: it drives where ERW is deployable. Life-cycle CO₂ of grinding, transport, and spreading is separately deducted from gross removal.

ASSUMPTION 2 · CARBON PRICE

The carbon return uses a net VCM price.

$R_{CDR} = \text{durable CDR} \times (p - m)$

p = VCM carbon price (\$/tCO₂) · m = per-tonne MRV cost

Baseline today's price is **\$150/tCO₂**, but we sweep a full price ladder. Price, MRV and delivered cost move the public envelope through a single ratio, $(p - m)/c$; the only other lever is durable removal per tonne (Results).

At \$150 the durable public envelope is **1.95 Mha** (11.1% of treated cropland); under a uniform-20 dose it widens to 3.95 Mha (12.1%), because the lime-requirement dose is CDR-minimal by construction.

Public envelope grows with price · equilibrium, targeted

\$/TCO ₂	PUBLIC MHA	% TREATED	DURABLE CDR (MT)
\$100	0.49	2.8%	2.2
\$150	1.95	11.1%	5.6
\$250	2.88	16.4%	7.0
\$350	8.35	47.5%	17.5
\$500	13.42	76.4%	23.3

ASSUMPTION 3 · NET-EXPORT CDR

Credit only carbon that is durably exported.

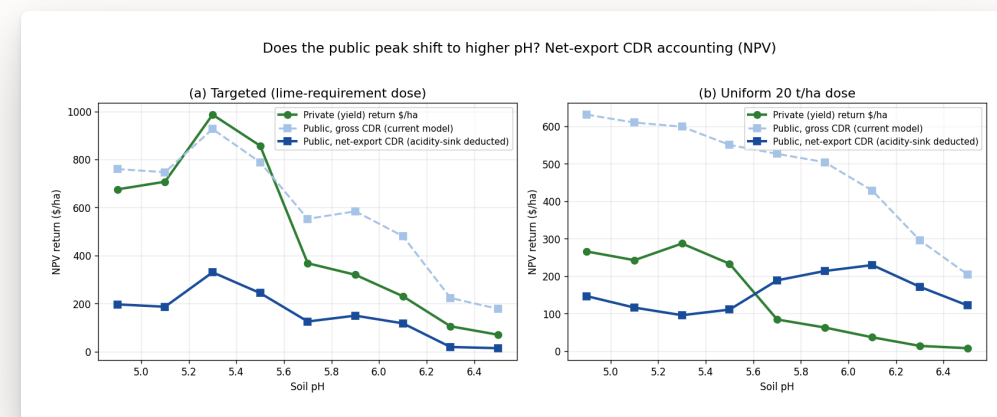
$$\text{durable CDR} = \max(\text{gross CDR} - F \cdot S, 0)$$

$F = 0.88 \text{ tCO}_2 \text{ per tCaCO}_3\text{-eq}$ · $S = \text{acidity sink (standing / maintenance)}$

Alkalinity spent neutralizing soil acidity re-releases its CO_2 , exactly as agricultural lime does, so it must be deducted. A first-order sequential bound.

WHY IT MATTERS

≈ **70%** of credited removal is cut at first application once acidity neutralization is netted out. Durable removal is largely a steady-state phenomenon (equilibrium retention 77–90%), which is why the headline uses the equilibrium regime.



ASSUMPTION 4 · OVER-LIMING PENALTY

Too much alkalinity can depress yield.

MECHANISM

Beyond the lime requirement, pH is pushed past the crop optimum; the EcoCrop response applies a yield haircut at the post-application pH.

BUFFERED SHIFT

pH change is damped by soil buffering ($\beta = k_{\beta} \cdot \text{ECEC}$); reacted alkalinity $A = \text{gross CDR} / F$.

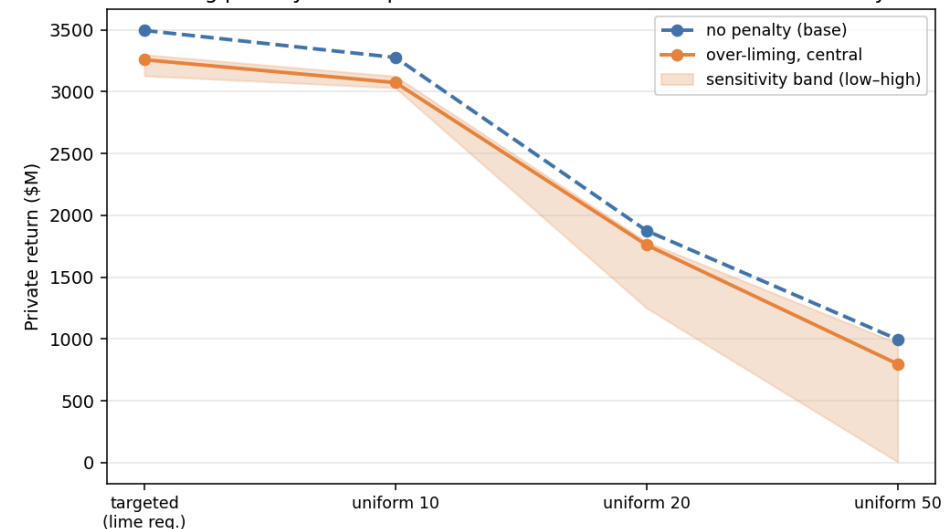
EFFECT

Concentrated on high uniform doses. Central estimate trims the targeted private return only modestly (~7%, band 6–10%); at uniform-50 the central hit is 20% with a 3–99% band.

SENSITIVITY

A low–high band over (k_{β} , realized-alkalinity) is carried; the uniform-50 case is where it bites hardest.

Over-liming penalty on the private return: central estimate and sensitivity band



ASSUMPTION 5 · KINETIC SATURATION

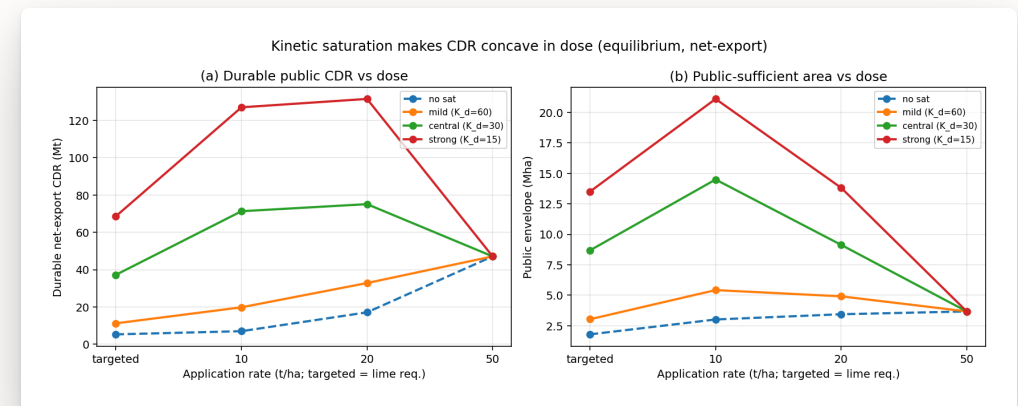
Weathering reacts less-than-linearly with dose.

$$\text{sat}(D) = \frac{K_d + 50}{K_d + D}$$

Effective reacted tonnage saturates in applied dose D , normalized to the Lewis (2021) 50 t/ha reference (sat = 1 there). Reported as a band over K_d .

KEY FINDING

Saturation makes CDR concave in dose (giving an interior optimum), but it does not drive the public envelope to zero. Below the 50 t/ha reference, per-tonne reactivity is higher, so at targeted doses durable CDR actually rises. What binds the carbon case is **MAC > price**, not kinetics.



DESIGN SPACE

Temporal regimes × allocation rules.

Two axes are swept alongside the assumption sets; the headline is **equilibrium, net-export**: targeted leads the private and intersection geography, uniform-20 the carbon case. Kinetics: Arrhenius ($E_a = 68.8 \text{ kJ/mol}$) · pH optimum 6.0 (2026-09 basis).

TEMPORAL REGIMES

YEAR-1

One-time dose & cost; durable CDR net of standing acidity. Most conservative on carbon.

NPV @ 10%

Discounted stream of relief and durable removal.

EQUILIBRIUM

Steady-state durable removal, the headline regime.

ALLOCATION RULES

TARGETED

LiTAS lime-equivalent dose, agronomically matched. Headline.

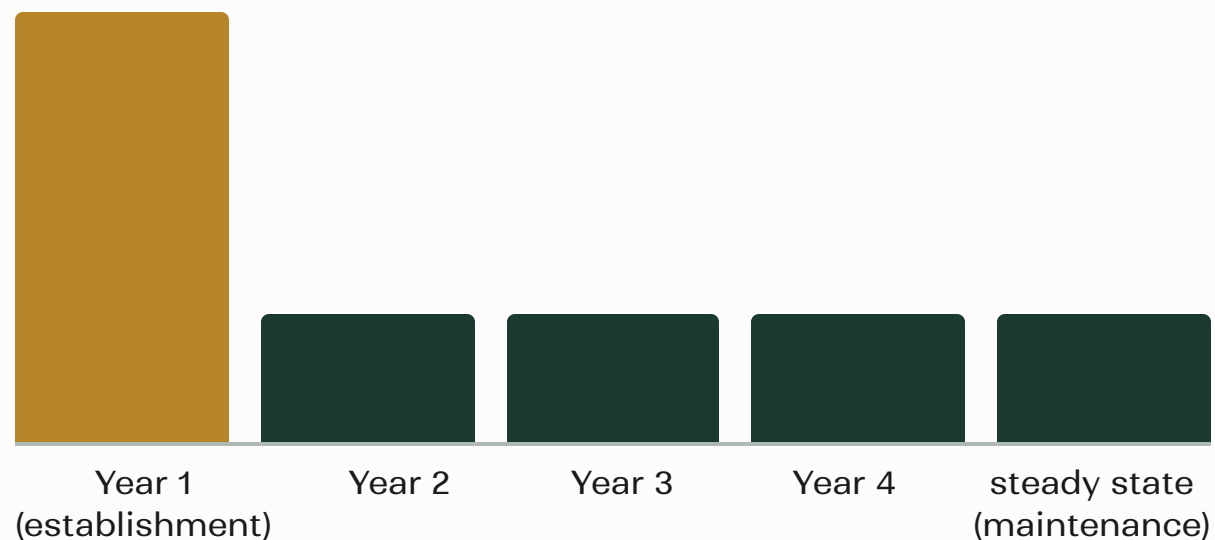
UNIFORM 10 / 20 / 50 T/HA

Fixed rates, tracing how CDR and over-liming scale with dose.

HOW TO READ THE RETURNS

Establishment dose vs. steady-state maintenance.

Full annual yield benefit is realized every year once soils are held at target.



Bar height $\propto$ basalt applied that period, charged at its full delivered cost.

EQUILIBRIUM: THE HEADLINE

Full annual yield vs the fully-costed maintenance top-up, once soils are at target. Private envelope: $\approx$ \$3.5 B yield for $\approx$ \$1.74 B rock ($\sim$ 2:1 per cycle). Excludes the establishment dose.

YEAR 1: ESTABLISHMENT

One-time full lime-equivalent dose: larger upfront cost, and durable carbon largely collapses (alkalinity first pays down the standing acidity debt). Reported separately as the establishment / upper-bound contrast.

Headline figures are steady-state annual: full yield against a fully-costed recurring dose. The upfront establishment investment is captured in the year-1 regime, not folded into the equilibrium ratio.

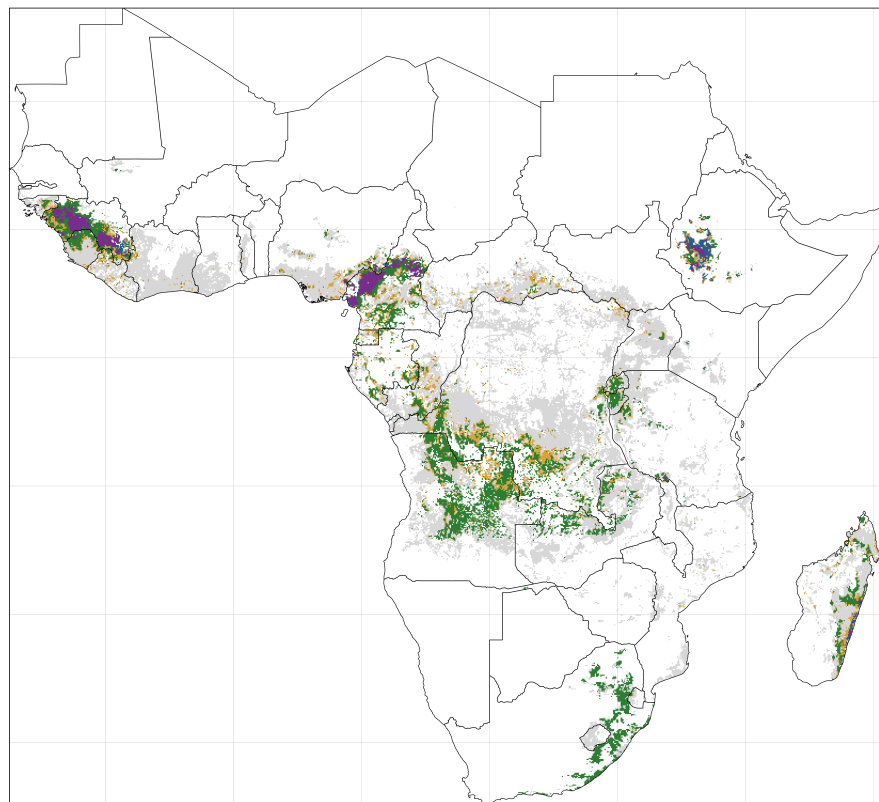
PART II

Results

Envelopes and the combined-only band · durable carbon and marginal abatement cost · what shifts the public frontier · optimal rates · the robust core geography.

A five-way pixel classification.

Return typology · equilibrium, net-export CDR, targeted



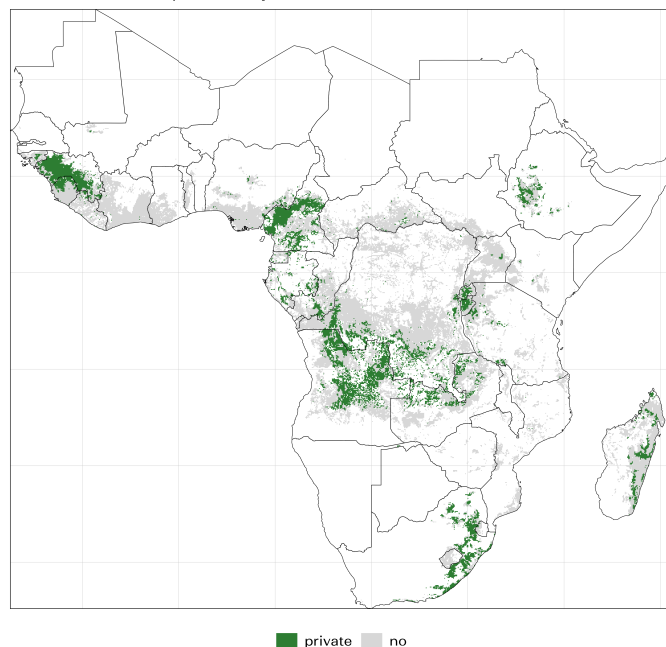
both private only public only combined only neither

- **Private-sufficient** · $R_{\text{agro}} - C > 0$
- **Public-sufficient** · $R_{\text{CDR}} - C > 0$
- **Intersection** · both hold at the same pixel
- **Combined-only** · neither alone, but $GM > 0$
- **Neither** · $GM \leq 0$, not deployable

RESULTS • PRIVATE ENVELOPE

Where yield alone pays.

Private-sufficient (yield) • equilibrium



Land where the agronomic return covers the full delivered cost with no carbon credit at all.

4.3 Mha

private-sufficient area

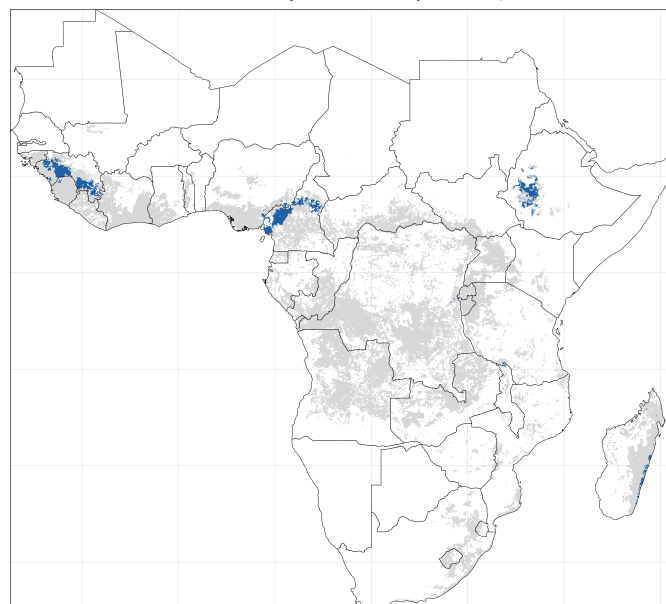
\$3.5 B

private return

The largest single envelope under the targeted dose. But it is a minority of acid cropland: most viable land needs more than yield to pencil.

RESULTS • PUBLIC ENVELOPE

Where carbon alone pays (durably).

Public-sufficient (durable net-export CDR) • equilibrium, \$150/tCO₂

■ public ■ no

Land where durable, net-export CDR at \$150/tCO₂ covers cost with no yield benefit.

1.95 Mha

durable public area (targeted)

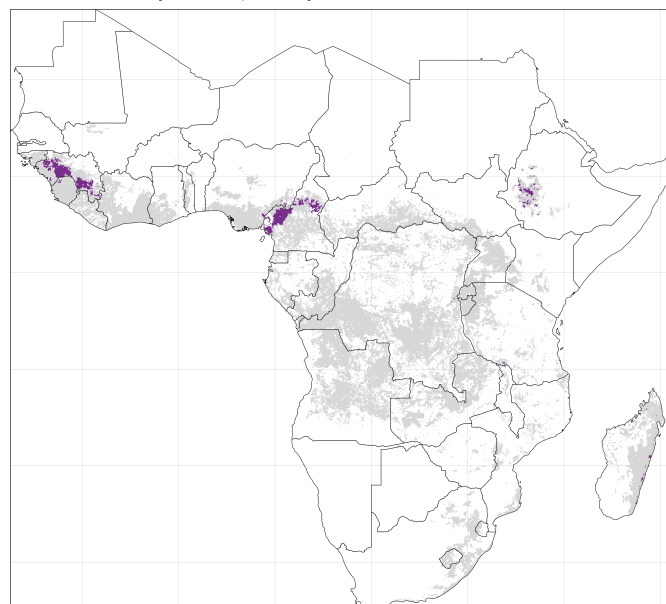
11.1 %of treated cropland • 3.95 Mha (12.1%)
under uniform-20

Smaller than the private envelope, and price-conditional. Net-export accounting and today's VCM price keep the carbon-only case tightly bound, the reason the combined case matters.

RESULTS • THE CENTRAL RESULT

The combined-only band is the policy object.

Intersection • equilibrium, net-export CDR



■ intersection ■ outside

2.7 MHA • COMBINED-ONLY

Neither the yield return nor the carbon return alone covers cost, yet together the margin is positive.

4.3

private (Mha)

2.0

public (Mha)

7.6

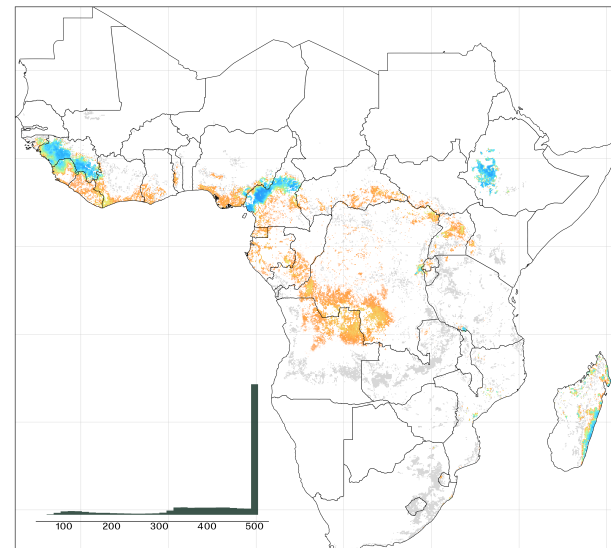
combined (Mha)

The economic point: voluntary carbon finance is not a bonus on an already-profitable practice; it is the marginal revenue that unlocks an otherwise-unprofitable, yield-improving investment.

RESULTS • WHAT BINDS THE CARBON CASE

Marginal abatement cost, not kinetics.

Marginal abatement cost • durable net-export CDR (equilibrium, targeted)



\$/tCO₂ (cap 500) 0 100 200 300 400 500

gray: treated cropland with no durable net-export CDR

\$330/tCO₂

area-weighted median MAC (targeted; \$276 under uniform-20), at the upper edge of published cost syntheses and well above today's \$150 VCM price

MAC = delivered cost per tonne of rock ÷ durable CDR per tonne. The public envelope is exactly the land where **MAC < net price** ($p - m$).

So the carbon case is limited by how much CO₂ a tonne of rock verifiably removes, the largest and least-resolved lever, and the clear field-calibration priority.

RESULTS • THREE LEVERS

What shifts the public envelope outward.

1 • HIGHER VCM PRICE (P)

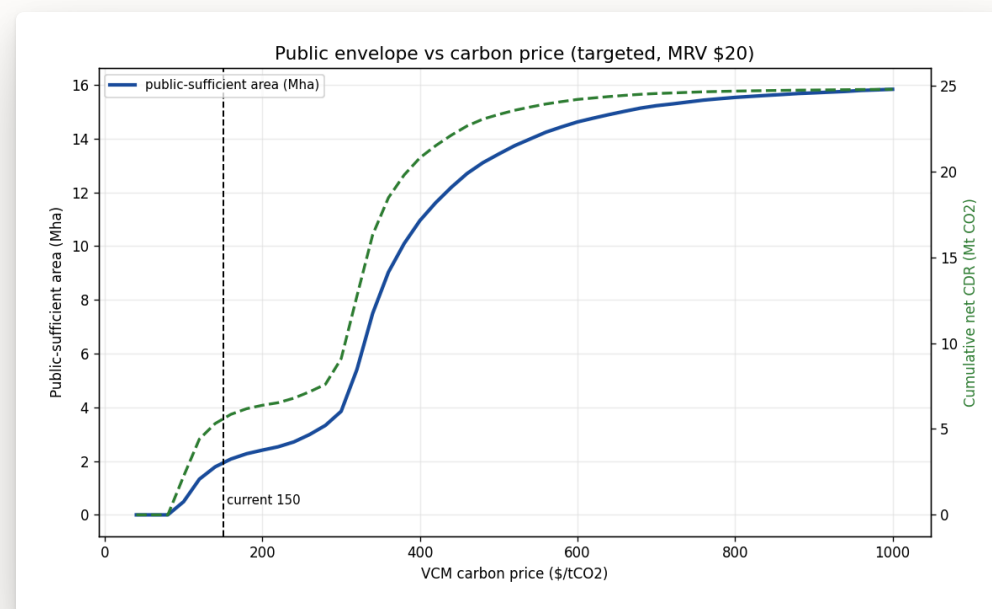
Public area climbs from 2.8% of treated cropland at \$100 to 76% at \$500. The most direct lever.

2 • LOWER DELIVERED COST (C)

Transport-cost reductions (e.g. solar haulage) lower MAC pixel-by-pixel; they act through the same $(p - m)/c$ ratio as price. See Assumption 1.

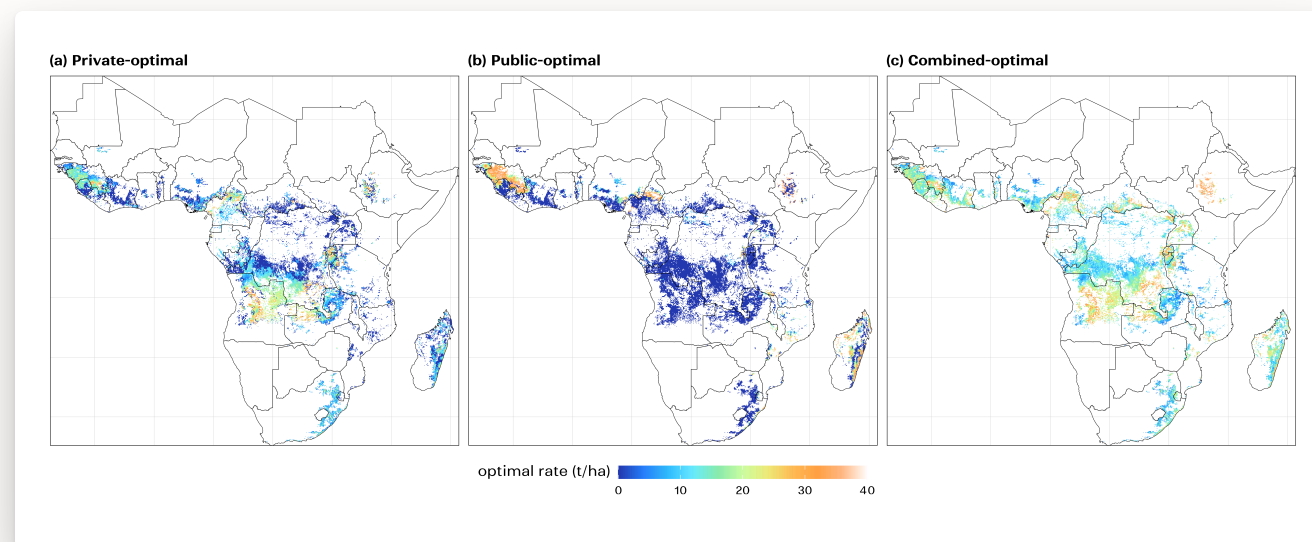
3 • MORE DURABLE CDR PER TONNE

Raising verified removal per tonne cuts MAC directly, and super-linearly: doubling the rate widens the uniform-20 public envelope ~6×. Kinetic saturation is second-order here.



RESULTS • OPTIMAL APPLICATION RATES

Each objective has an interior optimum.

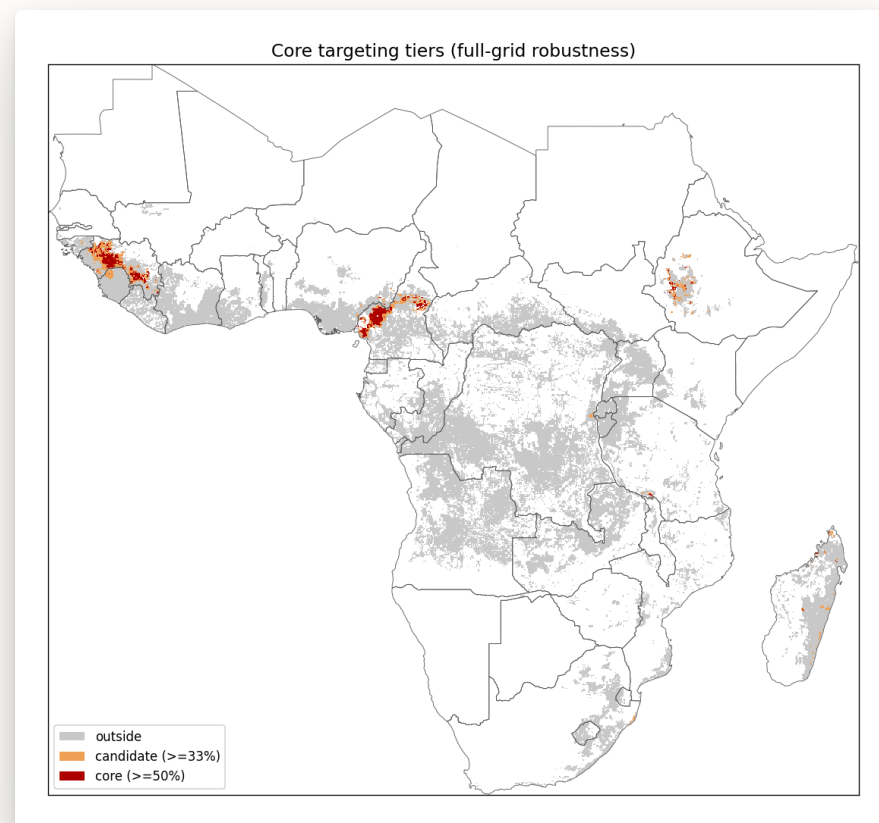


With both refinements, each per-pixel objective is concave, so a pixel-specific optimal rate exists for the private, public, and combined margins.

The combined optimum is **moderate** (typically well below blanket-high uniform doses): enough over-application to earn extra carbon, but not so much that over-liming and saturation erode the return.

RESULTS • ROBUST CORE

A defensible priority geography.



Intersecting the envelope across the plausible parameter grid isolates land that stays viable under most assumptions.

0.65 Mha

robust core-priority cropland
(candidate tier 1.22)

0.38 M

farms reached

~\$0.60 B

crop value at stake

This is the geography one would deploy first, and the base to expand as price rises or cost falls.

TAKEAWAYS

ERW is a coupled agronomic-and-carbon proposition.

1

The two returns rarely each suffice alone

But together they make a meaningful area of SSA cropland profitable that neither would, the 2.7 Mha combined-only band.

2

Carbon finance is the margin, not a bonus

On most viable land, VCM revenue is what unlocks a yield-improving investment. That reframes ERW away from a pure CDR supply curve.

3

The binding constraint is durable CDR per tonne

Net-export accounting cuts first-application removal ~70%; the area-weighted median MAC is \$330 (targeted) / \$276 (uniform-20). The price-cost ratio and verified removal per tonne are the two structural levers.

4

Treating it as coupled changes the answer

It changes both where to deploy first (the robust core) and what to invest in to expand it (field calibration of removal).

NEXT

Caveats and where we go next.

CAVEATS

Priorities

Field calibration of durable removal per tonne is the single highest-value input; it sets MAC and the whole public case.

Not yet modeled

Tenure and who finances the upfront cost, compaction / N₂O effects, and country-level credit eligibility (Article 6).

Farm counts

Based on national average holding sizes; order-of-magnitude.

NEXT

Registry-grade accounting

Align the net-export bound with VCM methodology counterfactuals: liming substitution, financial-additionality screens.

Optimal-rate maps

Fold the pixel-specific private and combined optimal-rate maps into the manuscript.

Share & collaborate

The repo is on GitHub; the interactive envelope explorer reproduces every lever of this analysis in one self-contained HTML file.



DISCUSSION

Thank you for your interest!

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